1. The primary responsibility of the Network Layer (Layer 3) in the OSI and TCP/IP models is:

(A) Host-to-host packet delivery and logical routing across multiple interconnected networks
(B) Framing bits into bytes
(C) Process-to-process port delivery
(D) Physical bit modulation

2. An Internet Protocol Version 4 (IPv4) address consists of how many binary bits?

(A) 32 bits (4 octets)
(B) 64 bits
(C) 128 bits
(D) 48 bits

3. IPv4 addresses are universally written in human-readable notation using:

(A) Dotted-decimal notation (e.g., 192.168.1.1)
(B) Hexadecimal notation
(C) Colon-separated octets
(D) Binary strings

4. In classical classful IPv4 addressing, an IP address whose first octet ranges from 1 to 126 belongs to:

(A) Class A
(B) Class B
(C) Class C
(D) Class D

5. In classful IPv4 addressing, a Class B network uses which default subnet mask?

(A) 255.255.0.0 (/16)
(B) 255.0.0.0 (/8)
(C) 255.255.255.0 (/24)
(D) 255.255.255.255 (/32)

6. In classful IPv4 addressing, an address with first octet ranging from 192 to 223 belongs to:

(A) Class C (default mask 255.255.255.0)
(B) Class A
(C) Class B
(D) Class D

7. IPv4 Class D addresses (first octet 224 to 239) are specifically reserved for:

(A) Multicasting
(B) Broadcasting to local subnets
(C) Loopback testing
(D) Private residential networks

8. The special IPv4 loopback address range reserved for internal software diagnostics and testing on the local host is:

(A) 127.0.0.0/8 (commonly 127.0.0.1)
(B) 0.0.0.0
(C) 192.168.0.1
(D) 255.255.255.255

9. Which RFC document specifies the globally reserved private IPv4 address blocks (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)?

(A) RFC 1918
(B) RFC 791
(C) RFC 826
(D) RFC 2460

10. Private IPv4 addresses cannot be directly routed across the public Internet; they require translation via:

(A) Network Address Translation (NAT / NAPT)
(B) DNS Forwarding
(C) ARP Broadcasting
(D) ICMP Tunneling

11. Classless Inter-Domain Routing (CIDR, RFC 1519) replaced rigid classful addressing by introducing:

(A) Variable-length prefix masking (/n notation, e.g., 192.168.1.0/24)
(B) 48-bit addresses
(C) Hexadecimal formatting
(D) Hardware-encoded IP routing

12. In CIDR notation 172.16.0.0/20, what does the suffix '/20' represent?

(A) The 20 most significant bits represent the network prefix, leaving 12 bits for host addresses
(B) There are 20 host computers
(C) The subnet mask is 255.255.255.0
(D) The router port is 20

13. How many total usable host IP addresses are available in a standard /24 IPv4 subnet?

(A) 254 usable hosts (2^8 - 2 = 256 - 2)
(B) 256 hosts
(C) 128 hosts
(D) 512 hosts

14. Why are two IP addresses subtracted when calculating usable host addresses in any IPv4 subnet?

(A) The first address is the Network ID and the last address is the Directed Broadcast Address
(B) One is for the router and one for DNS
(C) One is loopback and one is gateway
(D) Reserved for switch management

15. The minimum header length of an IPv4 datagram without optional fields is:

(A) 20 Bytes (160 bits)
(B) 40 Bytes
(C) 14 Bytes
(D) 64 Bytes

16. The Time-to-Live (TTL) field in the IPv4 header is decremented by 1 at every intermediate router to prevent:

(A) Packets from circulating endlessly in routing loops
(B) Buffer overflows
(C) Data corruption
(D) Unencrypted packet snooping

17. When an IPv4 packet's TTL field reaches zero, the router discards the packet and sends which ICMP message back to the source?

(A) ICMP Time Exceeded (Type 11)
(B) ICMP Destination Unreachable
(C) ICMP Echo Reply
(D) ICMP Redirect

18. Which IPv4 header field identifies the upper-layer transport protocol payload (e.g., value 6 for TCP, 17 for UDP)?

(A) Protocol Field
(B) Type of Service
(C) Identification
(D) Header Checksum

19. When an IP datagram exceeds the Maximum Transmission Unit (MTU) of an outgoing link, the router must perform:

(A) Packet Fragmentation
(B) Packet Compression
(C) Traffic Shaping
(D) Immediate dropping without notice

20. Which three IPv4 header fields coordinate packet fragmentation and reassembly at the destination?

(A) Identification, Flags (DF, MF), and Fragment Offset
(B) Source IP, Destination IP, and TTL
(C) Version, IHL, and Checksum
(D) Sequence Number, ACK, and Window

21. In the IPv4 Flags field, setting the 'Don't Fragment' (DF) bit instructs routers to:

(A) Never fragment the datagram; if it exceeds link MTU, drop it and send an ICMP error
(B) Allow unlimited fragmentation
(C) Fragment only at destination
(D) Compress payload before fragmenting

22. In IPv4 fragmentation, the Fragment Offset value is measured in units of:

(A) 8-byte blocks (64 bits)
(B) 1 byte
(C) 16 bytes
(D) 512 bytes

23. Internet Protocol Version 6 (IPv6) was designed to resolve IPv4 address exhaustion, expanding address length to:

(A) 128 bits (16 bytes)
(B) 64 bits
(C) 256 bits
(D) 48 bits

24. An IPv6 address is conventionally represented in human-readable notation as:

(A) 8 groups of 4 hexadecimal digits separated by colons (e.g., 2001:0db8:85a3::8a2e:0370:7334)
(B) 4 decimal numbers separated by dots
(C) 16 decimal bytes
(D) 32 binary nibbles

25. In IPv6 shorthand formatting, a single contiguous run of consecutive all-zero 16-bit blocks can be compressed using:

(A) A double colon (::), permitted at most once in an address
(B) Three hyphens (---)
(C) An asterisk (*)
(D) A slash (/)

26. The standard fixed base header length of an IPv6 packet is:

(A) 40 Bytes
(B) 20 Bytes
(C) 64 Bytes
(D) 32 Bytes

27. Which field present in IPv4 was completely REMOVED from the IPv6 base header to accelerate router processing speed?

(A) Header Checksum (error checking is deferred to Layer 2 and Layer 4)
(B) Hop Limit
(C) Source Address
(D) Payload Length

28. In IPv6, the IPv4 'Time to Live' (TTL) field was renamed to the more descriptive name:

(A) Hop Limit
(B) Packet Lifetime
(C) Router Countdown
(D) Drop Counter

29. Which address type supported in IPv4 was completely ELIMINATED in IPv6 and replaced by efficient Multicast groups?

(A) Broadcast Addressing
(B) Unicast Addressing
(C) Anycast Addressing
(D) Loopback Addressing

30. The special IPv6 loopback address is written in compressed notation as:

(A) ::1 (equivalent to 127.0.0.1 in IPv4)
(B) ::0
(C) fe80::1
(D) ff02::1

31. The process of forwarding packets hop-by-hop from a source network to a destination network based on routing tables is:

(A) Routing
(B) Switching
(C) Bridging
(D) Multiplexing

32. A Routing Table on a router stores which core mapping entries?

(A) Destination Network Prefix, Subnet Mask, Next-Hop IP Gateway, and Outgoing Interface
(B) MAC addresses and switch ports
(C) URLs and IP addresses
(D) Port numbers and application names

33. Static Routing differs from Dynamic Routing because static routes:

(A) Are manually configured by network administrators and do not adapt automatically to topology failures
(B) Update automatically via routing protocols
(C) Use Dijkstra's algorithm dynamically
(D) Require high CPU overhead

34. An Autonomous System (AS) in Internet routing is defined as:

(A) A collection of connected IP routing networks under the control of a single administrative entity running a common routing policy
(B) A standalone computer running Linux
(C) A root DNS server cluster
(D) A local home Wi-Fi network

35. Routing protocols operating strictly WITHIN an individual Autonomous System (e.g., OSPF, RIP, EIGRP) are classified as:

(A) Interior Gateway Protocols (IGP)
(B) Exterior Gateway Protocols (EGP)
(C) Border Gateway Protocols
(D) Mesh Routing Protocols

36. The primary de facto Exterior Gateway Protocol (EGP) used to route traffic BETWEEN different Autonomous Systems across the global Internet is:

(A) Border Gateway Protocol (BGP-4)
(B) OSPF
(C) RIP
(D) IS-IS

37. Routing Information Protocol (RIP) is a distance-vector routing protocol that uses which metric to determine path cost?

(A) Hop Count (maximum metric capped at 15 hops; 16 represents infinity/unreachable)
(B) Link bandwidth
(C) Cumulative delay
(D) Monetary link cost

38. Open Shortest Path First (OSPF) is an advanced link-state routing protocol based on which graph algorithm?

(A) Dijkstra's Shortest Path First (SPF) Algorithm
(B) Bellman-Ford Algorithm
(C) Floyd-Warshall Algorithm
(D) Kruskal's Algorithm

39. In OSPF, the network metric used to calculate path cost is inversely proportional to:

(A) Interface Bandwidth (Cost = Reference_Bandwidth / Interface_Bandwidth)
(B) Hop count
(C) Packet error rate
(D) Physical cable length

40. Internet Control Message Protocol (ICMP) is a Network layer companion protocol used primarily for:

(A) Network diagnostics, error reporting, and operational query messages (e.g., Echo Request/Reply in 'ping')
(B) Transferring web pages
(C) Encrypted file transfer
(D) Resolving domain names

41. What is the network ID and broadcast address for the host IP 192.168.10.68 with subnet mask 255.255.255.224 (/27)?

(A) Network ID: 192.168.10.64; Broadcast: 192.168.10.95
(B) Network ID: 192.168.10.0; Broadcast: 192.168.10.255
(C) Network ID: 192.168.10.64; Broadcast: 192.168.10.127
(D) Network ID: 192.168.10.32; Broadcast: 192.168.10.63

42. How many subnets and usable hosts per subnet are created by subnetting a Class C network (/24) using a /28 mask?

(A) 16 subnets with 14 usable hosts each (2^4 = 16 subnets; 2^4 - 2 = 14 hosts)
(B) 8 subnets with 30 hosts
(C) 4 subnets with 62 hosts
(D) 32 subnets with 6 hosts

43. A company requires 50 usable host IP addresses on a single subnet. What is the most efficient CIDR prefix length?

(A) /26 (provides 2^(32 - 26) - 2 = 64 - 2 = 62 usable addresses)
(B) /27 (only 30 hosts)
(C) /25 (126 hosts - wasteful)
(D) /28 (14 hosts)

44. What is the result of supernetting (aggregating) the four contiguous /24 networks: 192.168.0.0/24, 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24?

(A) 192.168.0.0/22 (Mask: 255.255.252.0)
(B) 192.168.0.0/21
(C) 192.168.0.0/23
(D) 192.168.0.0/16

45. In router forwarding, if multiple matching route entries exist in the routing table for a destination IP, the router selects the route based on:

(A) Longest Prefix Match (the route with the most specific, longest subnet mask)
(B) First matching entry in table
(C) Shortest subnet mask
(D) Lowest metric regardless of mask

46. An IPv4 datagram of total length 4000 bytes (20 bytes header + 3980 bytes payload) arrives at an outgoing link with MTU = 1500 bytes. How many fragments are created?

(A) 3 fragments (Fragment 1: 1480 payload + 20 hdr; Fragment 2: 1480 payload + 20 hdr; Fragment 3: 1020 payload + 20 hdr)
(B) 2 fragments
(C) 4 fragments
(D) 1 fragment

47. For the fragments generated in Q46, what are the Fragment Offset values for Fragment 1, Fragment 2, and Fragment 3?

(A) Fragment 1: 0; Fragment 2: 185 (1480 / 8); Fragment 3: 370 (2960 / 8)
(B) 0, 1480, 2960
(C) 0, 1, 2
(D) 0, 185, 500

48. For the fragments in Q46, what are the values of the More Fragments (MF) flag for Fragment 1, Fragment 2, and Fragment 3?

(A) Fragment 1: MF = 1; Fragment 2: MF = 1; Fragment 3: MF = 0
(B) All MF = 1
(C) All MF = 0
(D) Fragment 1: MF = 0; Others = 1

49. Path MTU Discovery (PMTUD, RFC 1191) determines the minimum MTU along an end-to-end network path by:

(A) Transmitting IP packets with the Don't Fragment (DF) bit set and listening for ICMP 'Fragmentation Needed' replies
(B) Probing with traceroute
(C) Reading DNS records
(D) Querying ARP servers

50. In Network Address Translation (NAT), Port Address Translation (PAT / NAPT) allows thousands of internal hosts to share a single public IP by:

(A) Mapping internal socket pairs (Private IP + Private Port) to distinct source port numbers on the single Public IP address
(B) Rotating IP addresses every second
(C) Compressing IP headers
(D) Using IPv6 tunneling

51. The 'Count-to-Infinity Problem' in Distance Vector routing protocols (e.g., RIP) occurs due to:

(A) Slow propagation of route failure news, causing neighboring routers to circulate outdated metrics incrementing up to infinity
(B) Routers running out of memory
(C) Hardware loopback shorts
(D) Packet collision storms

52. Which algorithmic technique helps mitigate the count-to-infinity problem by prohibiting a router from advertising a route back out the same interface it learned it from?

(A) Split Horizon
(B) Poison Reverse
(C) Link State flooding
(D) Longest prefix match

53. Split Horizon with Poison Reverse improves basic Split Horizon by:

(A) Explicitly advertising the route back out the incoming interface with an infinite metric (cost 16 in RIP) to immediately invalidate it
(B) Dropping all routing updates
(C) Doubling hello timers
(D) Using static routes

54. In OSPF, routers exchange their local link states with all other routers in the area using:

(A) Link State Advertisements (LSAs) flooded via reliable multicast
(B) Distance vector tables
(C) BGP update messages
(D) ICMP redirect frames

55. Every router participating in an OSPF area possesses an IDENTICAL copy of the:

(A) Link State Database (LSDB), representing the complete network topology graph
(B) Routing table
(C) ARP cache
(D) DNS table

56. In multi-access broadcast networks (e.g., Ethernet), OSPF reduces LSA flooding overhead by electing a:

(A) Designated Router (DR) and Backup Designated Router (BDR)
(B) Root Bridge
(C) Master Gateway
(D) Default Route Provider

57. OSPF Area 0 is a mandatory, core architectural component in any multi-area OSPF network known as the:

(A) Backbone Area (Area 0.0.0.0)
(B) Stub Area
(C) Totally Stubby Area
(D) Not-So-Stubby Area (NSSA)

58. An OSPF router that connects one or more regular non-backbone areas directly to the central Backbone Area 0 is an:

(A) Area Border Router (ABR)
(B) Autonomous System Boundary Router (ASBR)
(C) Internal Router
(D) Designated Router

59. An OSPF router that redistributes routes learned from external routing protocols (e.g., BGP, RIP, static routes) into OSPF is an:

(A) Autonomous System Boundary Router (ASBR)
(B) Area Border Router (ABR)
(C) Designated Router (DR)
(D) Switch Router

60. Border Gateway Protocol (BGP) is classified as which type of routing protocol?

(A) Path Vector Routing Protocol
(B) Distance Vector Protocol
(C) Link State Protocol
(D) Centralized Controller Protocol

61. BGP prevents inter-domain routing loops across Autonomous Systems by:

(A) Recording the entire sequence of Autonomous System Numbers (AS-Path attribute) in the route advertisement and discarding routes containing its own AS
(B) Using hop counts capped at 15
(C) Running Dijkstra's algorithm globally
(D) Flooding link states worldwide

62. BGP establishes reliable peering sessions between neighboring routers using which transport protocol and port number?

(A) TCP Port 179
(B) UDP Port 520
(C) IP Protocol 89
(D) TCP Port 80

63. Dynamic Host Configuration Protocol (DHCP) operates over UDP (Ports 67 and 68). What is the standard 4-step handshake sequence for client IP assignment?

(A) DORA: Discover (broadcast), Offer (unicast/broadcast), Request (broadcast), Acknowledgment (ACK)
(B) SYN, SYN-ACK, ACK, FIN
(C) Request, Reply, Confirm, Connect
(D) Probe, Allocate, Validate, Lease

64. When a client computer boots up without an IP configuration, its initial DHCP Discover broadcast uses which source and destination IP addresses?

(A) Source: 0.0.0.0; Destination: 255.255.255.255
(B) Source: 127.0.0.1; Destination: 192.168.1.1
(C) Source: 0.0.0.0; Destination: 192.168.1.254
(D) Source: 255.255.255.255; Destination: 0.0.0.0

65. What is the primary role of a DHCP Relay Agent configured on a local router interface?

(A) To forward broadcast DHCP Discover/Request messages from local clients as unicast IP packets to a remote DHCP server on a different subnet
(B) To generate random IP addresses
(C) To block DHCP attacks
(D) To assign DNS domain names

66. DHCP Snooping is a Layer 2 security mechanism on enterprise switches that prevents rogue DHCP server attacks by:

(A) Classifying switch ports as Trusted (facing legitimate DHCP servers) or Untrusted (clients) and dropping DHCP offers from untrusted ports
(B) Disabling DHCP entirely
(C) Filtering MAC addresses
(D) Encrypting DHCP packets

67. The diagnostic network utility 'traceroute' (or 'tracert' in Windows) maps the hop-by-hop router path to a destination by:

(A) Incrementing the IP TTL field sequentially (TTL = 1, 2, 3...) and listening for ICMP 'Time Exceeded' messages from intermediate routers
(B) Sending ARP requests across the Internet
(C) Querying DNS root servers
(D) Querying BGP looking glass servers

68. In IPv6, Address Resolution Protocol (ARP) was replaced by which ICMPv6-based protocol?

(A) Neighbor Discovery Protocol (NDP) using Neighbor Solicitation (NS) and Neighbor Advertisement (NA)
(B) Reverse ARP (RARP)
(C) DHCPv6 only
(D) IGMPv3

69. IPv6 Stateless Address Autoconfiguration (SLAAC) allows an IPv6 client to automatically generate its own globally unique address by:

(A) Combining a 64-bit network prefix received from a Router Advertisement (RA) with a 64-bit Interface Identifier (e.g., EUI-64 derived from MAC)
(B) Requesting an address from a stateful DHCP server
(C) Using random 32-bit hashes
(D) Hardcoding a static IP

70. In the EUI-64 process for generating an IPv6 interface ID from a 48-bit MAC address, what two modifications are performed?

(A) Inserting the 16-bit hex value 0xFFFE in the middle of the MAC and inverting the 7th bit (Universal/Local bit)
(B) Hashing the MAC with SHA-256
(C) Inverting all 48 bits
(D) Appending 64 zeros

71. IPv6 Link-Local addresses used strictly for communication within a single local physical link begin with which prefix?

(A) fe80::/10
(B) 2001::/16
(C) ff00::/8
(D) fc00::/7

72. IPv6 Unique Local Addresses (ULA), equivalent to IPv4 private addresses (RFC 1918) for local enterprise routing, reside in the range:

(A) fc00::/7 (typically fd00::/8)
(B) fe80::/10
(C) 2000::/3
(D) ::/128

73. Which transition mechanism allows IPv6 packets to be transported across an existing IPv4-only network infrastructure?

(A) IPv6 Tunneling (e.g., 6in4, 6to4, Teredo, GRE)
(B) Subnetting
(C) CIDR aggregation
(D) NAT64 translation only

74. Dual-Stack architecture for IPv4-to-IPv6 transition involves:

(A) Running IPv4 and IPv6 protocol stacks concurrently on all network hosts, operating systems, and routers
(B) Converting all IPv4 headers to IPv6 at the NIC
(C) Disabling IPv4 globally
(D) Using proxy servers exclusively

75. Virtual Router Redundancy Protocol (VRRP, RFC 5798) provides high network gateway availability by:

(A) Grouping two or more physical routers into a single Virtual Default Gateway sharing a virtual IP and virtual MAC address
(B) Distributing packets across two switches
(C) Running OSPF and RIP simultaneously
(D) Doubling router memory

76. An ISP owns the address block 200.100.0.0/16. It needs to allocate equal-sized address blocks to 64 customer organizations. What is the CIDR prefix allocated to each customer and how many total addresses does each receive?

(A) 200.100.x.x/22, allocating 1024 addresses per customer (16 + log2(64) = 16 + 6 = /22; 2^10 = 1024)
(B) /24 allocating 256 addresses
(C) /20 allocating 4096 addresses
(D) /26 allocating 64 addresses

77. A router has four forwarding entries in its routing table: (1) 128.96.39.0/24 -> Port 1, (2) 128.96.39.128/25 -> Port 2, (3) 128.96.0.0/16 -> Port 3, (4) 0.0.0.0/0 -> Port 4. Out of which port will a packet destined for IP 128.96.39.150 be forwarded?

(A) Port 2 (Matching entries are 1, 2, 3, 4; Longest Prefix Match is /25 on Port 2)
(B) Port 1 (/24)
(C) Port 3 (/16)
(D) Port 4 (Default route)

78. For the router in Q77, out of which port will a packet destined for IP 128.96.39.10 be forwarded?

(A) Port 1 (Matches /24, /16, /0; Longest Prefix is /24; does not match /25 since 10 < 128)
(B) Port 2
(C) Port 3
(D) Port 4

79. In Bellman-Ford distributed distance-vector routing, the update equation for router x to update its distance d_x(y) to destination y is:

(A) d_x(y) = min_v { c(x, v) + d_v(y) }, where v iterates over all immediate neighbors of x
(B) d_x(y) = c(x, y)
(C) d_x(y) = d_x(v) + c(v, y)
(D) d_x(y) = max_v { c(x, v) }

80. In Dijkstra's algorithm for a graph with V nodes and E edges implemented using a min-heap priority queue, the computational time complexity is:

(A) O((V + E) * log(V))
(B) O(V^2)
(C) O(V * E)
(D) O(E^2)

81. An IP packet carrying a 64-byte ICMP payload has total size 84 bytes (20 bytes IPv4 header + 64 bytes ICMP). If sent over an ATM network with 53-byte cells (5 bytes header + 48 bytes payload using AAL5 with 8-byte trailer), how many ATM cells are generated?

(A) 2 ATM cells (Total AAL5 PDU = 84 + 8 = 92 bytes; ceil(92 / 48) = 2 cells)
(B) 1 cell
(C) 3 cells
(D) 4 cells

82. A NAT router has a single public IP address 203.0.113.5. If port numbers are 16-bit integers (excluding well-known ports 0-1023), what is the theoretical maximum number of concurrent translated outbound TCP connections supported?

(A) 64,512 concurrent connections (65,536 - 1024)
(B) 256 connections
(C) 1024 connections
(D) 1,048,576 connections

83. In an enterprise multi-homed BGP network, how does a network administrator influence outbound traffic exit selection across multiple external transit providers?

(A) By configuring the BGP Local Preference (LOCAL_PREF) attribute (higher local preference is preferred across the AS)
(B) By prepending AS numbers
(C) By adjusting the MED attribute
(D) By tuning OSPF cost

84. To influence INBOUND traffic entering an Autonomous System from an external peering partner, a BGP engineer adjusts the:

(A) Multi-Exit Discriminator (MED) attribute (lower metric is preferred by neighbor AS) or AS-Path Prepending
(B) Local Preference
(C) Weight attribute
(D) Router ID

85. AS-Path Prepending influences incoming BGP traffic by:

(A) Artificially lengthening the advertised AS-Path by repeating its own AS number multiple times, making the route appear less attractive to remote networks
(B) Encrypting the route
(C) Dropping BGP keepalives
(D) Setting next-hop to 0.0.0.0

86. In an OSPF network with 5 routers connected via a full mesh of point-to-point links, what is the total number of bidirectional adjacency links formed?

(A) 10 adjacencies (5 * 4 / 2)
(B) 20 adjacencies
(C) 5 adjacencies
(D) 25 adjacencies

87. In OSPF, what is the default reference bandwidth used to calculate interface cost (Cost = 10^8 / Bandwidth_bps)?

(A) 100 Mbps (100,000,000 bps; consequently 100 Mbps, 1 Gbps, and 10 Gbps links all yield cost = 1 unless auto-cost reference-bandwidth is reconfigured)
(B) 10 Mbps
(C) 1 Gbps
(D) 10 Gbps

88. In Equal-Cost Multi-Path (ECMP) routing, when a router possesses four identical-cost routes to a destination subnet, it balances traffic by:

(A) Hashing packet 5-tuples (Src IP, Dst IP, Protocol, Src Port, Dst Port) to forward individual flows deterministically across all 4 equal paths
(B) Round-robin bit-by-bit interleaving
(C) Duplicating all packets 4 times
(D) Sending all traffic to the first port

89. In IPv4 header checksum calculation, why must intermediate routers recompute the checksum at EVERY hop?

(A) Because the Time-to-Live (TTL) field is decremented by 1 at every router hop, modifying the 16-bit header contents
(B) Because the payload changes
(C) Because the source IP is updated
(D) Because packet length varies

90. How can a router verify the IPv4 header checksum in a single operation without separate subtraction?

(A) By computing the 16-bit 1's complement sum over the ENTIRE 20-byte header INCLUDING the checksum field; the result must equal 0xFFFF (or 0x0000)
(B) By checking the parity bit
(C) By performing CRC-32 division
(D) By validating against DNS

91. In Multicast Listener Discovery (MLD) for IPv6, what multicast address corresponds to the 'All Routers' link-local group?

(A) ff02::2 (equivalent to 224.0.0.2 in IPv4)
(B) ff02::1
(C) ff01::1
(D) ff05::2

92. The Solicited-Node Multicast Address in IPv6 used for address resolution in NDP is formed by appending the low-order 24 bits of the target IPv6 address to:

(A) ff02::1:ff00:0/104
(B) ff02::1
(C) fe80::/10
(D) ff05::/16

93. Reverse Path Forwarding (RPF) is an essential loop-prevention algorithm used in Multicast Routing (PIM) that accepts a multicast packet ONLY if:

(A) It arrives on the specific interface that the router would use to unicast back to the multicast source
(B) It contains a valid TCP header
(C) The destination IP is in the routing table
(D) The TTL exceeds 50

94. In Protocol Independent Multicast - Sparse Mode (PIM-SM), multicast sources initially send data to which central rendezvous node?

(A) Rendezvous Point (RP)
(B) Root Bridge
(C) BGP Speaker
(D) Designated Switch

95. Multiprotocol Label Switching (MPLS) accelerates packet forwarding across core service provider backbones by:

(A) Attaching short, fixed-length 32-bit labels (MPLS shims) to packets and forwarding via Label Switched Paths (LSP) without inspecting IP headers
(B) Replacing IP addresses with MAC addresses
(C) Compressing payload data with GZIP
(D) Using circuit switching lines

96. An MPLS Label Stack Entry (Shim Header) is 32 bits long, consisting of: 20-bit Label, 3-bit Traffic Class (EXP/QoS), 1-bit Bottom of Stack (S), and:

(A) 8-bit Time-to-Live (TTL)
(B) 4-bit Checksum
(C) 16-bit Length
(D) 8-bit Protocol ID

97. Software-Defined Networking (SDN) fundamentally decouples network architecture by separating:

(A) The Control Plane (centralized routing intelligence/controller) from the Data Plane (fast packet forwarding hardware)
(B) The Client from the Server
(C) Copper lines from fiber lines
(D) IPv4 from IPv6

98. OpenFlow is the standardized communications protocol that enables an SDN Controller to directly program the:

(A) Flow forwarding tables in the data plane switches
(B) CPU cache coherency
(C) BGP autonomous system policies
(D) DNS root database

99. Segment Routing (SR) modernizes packet forwarding in service provider networks by:

(A) Encoding the network forwarding path directly in the packet header as an ordered list of instructions (Segments/MPLS labels or IPv6 SIDs) without maintaining per-flow core state

(B) Requiring RSVP-TE signaling on every router

(C) Running RIP across all links

(D) Disabling IP routing tables

100. Network Function Virtualization (NFV) replaces dedicated proprietary network hardware appliances (firewalls, load balancers, routers) with:

(A) Virtual Network Functions (VNFs) running as software virtual machines or containers on commercial off-the-shelf (COTS) standard x86 servers

(B) Hardwired ASIC circuits

(C) Mainframe computing clusters

(D) Analog electronic filter banks

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the architecture of IPv4 Addressing. Detail Classful Addressing (Classes A through E), default subnet masks, and address ranges. Explain the transition to Classless Inter-Domain Routing (CIDR).

Q2. Explain IPv4 Subnetting. An organization is allocated the network address 192.168.1.0/24 and needs to create 4 equal subnets. Calculate for each subnet: (1) Subnet mask, (2) Subnet Network ID, (3) Range of usable host IP addresses, (4) Subnet Broadcast address.

Q3. Explain IPv4 Packet Fragmentation. Detail the roles of the Identification, Flags (DF, MF), and Fragment Offset fields. An IP datagram with 4000 bytes total length (20 bytes header + 3980 bytes payload) passes through a link with MTU = 1500 bytes. Show the mathematical values for all

Q4. Compare IPv4 and IPv6 architectures. Detail the improvements introduced in IPv6 regarding: Address space, Header format simplification, Security, Quality of Service, Autoconfiguration, and Elimination of broadcasts.

Q5. Compare Distance Vector Routing (Bellman-Ford Algorithm) with Link State Routing (Dijkstra's Algorithm). Detail their convergence speeds, routing metrics, protocol message overhead, and vulnerability to routing loops.

Q6. Analyze Network Layer and routing protocol topic #6: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q7. Analyze Network Layer and routing protocol topic #7: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q8. Analyze Network Layer and routing protocol topic #8: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q9. Analyze Network Layer and routing protocol topic #9: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q10. Analyze Network Layer and routing protocol topic #10: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q11. Analyze Network Layer and routing protocol topic #11: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q12. Analyze Network Layer and routing protocol topic #12: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q13. Analyze Network Layer and routing protocol topic #13: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q14. Analyze Network Layer and routing protocol topic #14: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q15. Analyze Network Layer and routing protocol topic #15: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q16. Analyze Network Layer and routing protocol topic #16: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q17. Analyze Network Layer and routing protocol topic #17: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q18. Analyze Network Layer and routing protocol topic #18: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q19. Analyze Network Layer and routing protocol topic #19: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q20. Analyze Network Layer and routing protocol topic #20: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q21. Analyze Network Layer and routing protocol topic #21: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q22. Analyze Network Layer and routing protocol topic #22: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q23. Analyze Network Layer and routing protocol topic #23: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q24. Analyze Network Layer and routing protocol topic #24: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q25. Analyze Network Layer and routing protocol topic #25: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q26. Analyze Network Layer and routing protocol topic #26: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q27. Analyze Network Layer and routing protocol topic #27: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q28. Analyze Network Layer and routing protocol topic #28: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q29. Analyze Network Layer and routing protocol topic #29: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

Q30. Analyze Network Layer and routing protocol topic #30: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, and MPLS traffic engineering.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

1. Mathematical Principles of Subnetting:
Subnetting borrows bits from the Host portion of an IP address and assigns them to the Network portion,
logically partitioning a single physical network into multiple smaller, independent broadcast domains.
- Number of subnets created by borrowing 's' bits: Subnets = 2^s.
- Number of usable hosts per subnet with 'h' remaining host bits: Usable Hosts = 2^h - 2.

2. Step-by-Step Subnet Calculations for 192.168.1.0/24:
- Base Network: 192.168.1.0/24 (Default mask: 255.255.255.0; 8 host bits available).
- Required Subnets = 4 => 2^s = 4 => s = 2 borrowed bits.
- Remaining Host Bits: h = 8 - 2 = 6 bits.
- New Subnet Prefix: /24 + 2 = /26.
- Subnet Mask in Binary: 11111111.11111111.11111111.11000000.
- Subnet Mask in Decimal: 255.255.255.192.
- Block Size (Increment per subnet): 2^h = 2^6 = 64 addresses per block.
- Usable Hosts per Subnet: 2^6 - 2 = 64 - 2 = 62 usable hosts.

3. Detailed Subnet Allocation Table:
- Subnet 1:
* Network ID: 192.168.1.0 /26
* Usable Host IP Range: 192.168.1.1 to 192.168.1.62
* Directed Broadcast Address: 192.168.1.63
- Subnet 2:
* Network ID: 192.168.1.64 /26
* Usable Host IP Range: 192.168.1.65 to 192.168.1.126
* Directed Broadcast Address: 192.168.1.127
- Subnet 3:
* Network ID: 192.168.1.128 /26
* Usable Host IP Range: 192.168.1.129 to 192.168.1.190
* Directed Broadcast Address: 192.168.1.191
- Subnet 4:
* Network ID: 192.168.1.192 /26
* Usable Host IP Range: 192.168.1.193 to 192.168.1.254
* Directed Broadcast Address: 192.168.1.255

>>Solution to Question 3: Explain IPv4 Packet Fragmentation. Detail the roles of the Identification, Flags (DF, MF), and Fragment Offset fields. An IP datagram w

- Flags (3 bits):
* Bit 0: Reserved (must be 0).
* Bit 1 (DF - Don't Fragment): 0 = Fragmentation allowed; 1 = Do not fragment (drop packet and return ICMP error if MTU exceeded).
* Bit 2 (MF - More Fragments): 1 = More fragments follow; 0 = Last fragment (or unfragmented packet).
- Fragment Offset (13 bits): Indicates the relative starting position of this fragment's data payload within the original unfragmented payload,
measured in units of 8-byte blocks (64 bits). Hence, payload sizes of all fragments except the last must be exact multiples of 8.

3. Numerical Fragmentation Problem Solution:
- Original Packet: Total length = 4000 bytes (Header = 20 bytes, Payload = 3980 bytes). ID = 54321, DF = 0.
- Outgoing Link MTU = 1500 bytes. Maximum payload per fragment = 1500 - 20 = 1480 bytes (conveniently divisible by 8: 1480 / 8 = 185).
- Fragment 1:
* Payload length: 1480 bytes (bytes 0 to 1479).
* Total Length: 1480 + 20 = 1500 bytes.
* ID = 54321; DF = 0; MF = 1 (more fragments follow).
* Fragment Offset = 0 / 8 = 0.
- Fragment 2:
* Payload length: 1480 bytes (bytes 1480 to 2959).
* Total Length: 1480 + 20 = 1500 bytes.
* ID = 54321; DF = 0; MF = 1.
* Fragment Offset = 1480 / 8 = 185.
- Fragment 3 (Final Fragment):
* Remaining Payload: 3980 - (1480 + 1480) = 3980 - 2960 = 1020 bytes (bytes 2960 to 3979).
* Total Length: 1020 + 20 = 1040 bytes.
* ID = 54321; DF = 0; MF = 0 (Last fragment!).
* Fragment Offset = 2960 / 8 = 370.

>>Solution to Question 4: Compare IPv4 and IPv6 architectures. Detail the improvements introduced in IPv6 regarding: Address space, Header format simplificat

Advertisements and EUI-64 / random interface IDs.
True 'plug-and-play' connectivity without dedicated DHCP servers.
- 4. Elimination of Broadcasts:
* IPv4: Heavily dependent on disruptive broadcast traffic (255.255.255.255, ARP requests), which forces every host on the subnet to interrupt its CPU.
* IPv6: Zero broadcast addressing! All broadcast functions replaced by efficient Multicast groups (e.g., Solicited-Node Multicast in NDP) and Anycast.
- 5. Security Architecture:
* IPv4: IPsec was retrofitted as an optional add-on (RFC 4301).
* IPv6: Built from the ground up with native mandatory support for IPsec (AH and ESP extension headers).
- 6. Quality of Service (QoS):
* IPv4: 8-bit Type of Service (ToS) / DiffServ field.
* IPv6: 8-bit Traffic Class field plus a dedicated 20-bit Flow Label field that enables ultra-fast hardware routing of real-time multimedia flows without deep packet inspection.

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1. Theoretical Context:
In-depth technical analysis of network layer routing architecture and internetworking topic #6.

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #9: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal, a

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>> Solution to Question 10: Analyze Network Layer and routing protocol topic #10: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #13: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #17: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #20: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Solution Statement: Analyze Network Layer and routing protocol topic #23: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #26: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.

>>Problem Statement: Analyze Network Layer and routing protocol topic #29: Discuss BGP path vector attributes, ICMP diagnostic messaging, NAT traversal,

2. Mathematical & Algorithmic Formulation:
- Longest Prefix Match trie search complexity: O(W), where W = 32 or 128 bits.
- BGP path vector loop prevention algorithm.
- OSPF Dijkstra SPF tree computation.

3. Practical Internetworking Deployment:
Application in tier-1 global transit networks, software-defined WAN (SD-WAN), and multi-cloud transit gateways.
